

AWS S3 (Simple Storage Service)

00-foundations/s3

Overview

S3 is AWS object storage. It stores data as objects inside buckets and is designed for very high durability, massive scale, and broad integration with the rest of AWS.

In ML systems, S3 is usually the default data lake and artifact store. Raw data, curated datasets, model files, pipeline outputs, logs, and static assets often live there.

Core Concepts

- Buckets are top level containers and objects are individual files or blobs addressed by key. Prefixes are naming conventions that simulate folders and support partitioned layouts.
- Important features include versioning, lifecycle rules, server side encryption, bucket policies, replication, event notifications, and multiple storage classes.
- S3 provides strong read after write and list consistency, which simplifies many ingestion and pipeline patterns compared with older eventual consistency assumptions.

How It Works

- Applications upload or download objects through the S3 API, SDKs, CLI, or signed URLs. IAM and bucket policies decide who can access which paths.
- Data engineering jobs commonly read from raw prefixes, transform the data, and write curated outputs in columnar formats such as Parquet for analytics and downstream ML use.
- Many services integrate directly with S3, including Athena, Glue, SageMaker, Lambda event notifications, and CloudTrail log delivery.

AI and ML Relevance

- Training datasets are often staged in S3, and model artifacts are written back there after the training job completes.
- S3 can trigger downstream workflows when new data lands, which is useful for automated feature computation, validation, batch inference, or retraining pipelines.

Operational Notes

- Design the bucket key structure deliberately. Bad partitioning and inconsistent naming make analytics, retention, and lifecycle management painful later.
- Encrypt sensitive data, control public access at both the bucket and account level, and use lifecycle rules to move cold data to cheaper storage classes.
- Do not treat S3 like a filesystem with heavy small file chatter when a compact columnar layout would be more efficient and cheaper.

What To Remember

- S3 is object storage, not block storage and not a database.
- Bucket policy plus IAM policy plus encryption controls together define access posture.
- For ML, think of S3 as the durable backbone for data and artifacts rather than just a place to drop files.